INCH-POUND

OO-T-290F
28 February 1992
SUPERSEDING
OO-T-290E
20 May 1985

FEDERAL SPECIFICATION

THREADING MACHINES; PIPE AND BOLT, POWER DRIVEN (BENCH AND FLOOR TYPE)

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers pipe and bolt threading machines, bench and floor mounted.

1.2 Classification.

1.2.1 Types, classes, styles, and sizes. The machines covered by this specification shall be of the following types, classes, styles, and sizes, as specified (see 6.2):

Type I - Bench

Class 1 - Light duty

Style A - Stationary die-head

Size 2 - Threading range (min)
Pipe - 1/8 to 2 inches
Bolt - 1/4 to 2 inches

Style B - Revolving die-head

Size 2 - Threading range (min)
Pipe - 1/8 to 2 inches
Bolt - 1/4 to 2 inches

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-P, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3419

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Class 2 - Heavy duty

Style A - Stationary die-head

Size 2 - Threading range (min)
Pipe - 1/8 to 2 inches
Bolt - 1/4 to 2 inches

Type II - Floor

Class 1 - Light duty

Style A - Stationary die-head

Size 4 - Threading range (min)
Pipe - 1/2 to 4 inches
Bolt - 5/8 to 3 inches

Style B - Revolving die-head

Size 2 - Threading range (min)
Pipe - 1/4 to 2 inches
Bolt - 1/4 to 2 inches

Size 6 - Threading range (min)
Pipe - 1 to 6 inches
Bolt - 1 to 4 inches

Class 2 - Heavy duty

Style A - Stationary die-head

Size 2 - Threading range (min)
Pipe - 1/4 to 2 inches
Bolt - 1/4 to 2 inches

Size 4 - Threading range (min)
Pipe - 1/2 to 4 inches
Bolt - 3/8 to 3 inches

Size 6 - Threading range (min)
Pipe - 1 to 6 inches
Bolt - 1 to 4 inches

Size 8 - Threading range (min)
Pipe - 2-1/2 to 8 inches
Bolt - 2 to 6 inches

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2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
P-D-680	- Dry Cleaning Solvent
W-C-596	- Connector, Plug, Receptacle and Cable Outlet, - Electrical Power General, Specification for
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-1055	- Barrier Material, Waterproof, Flexible
PPP-C-843	- Cushioning Material, Cellulosic
PPP-D-705	- Drum, Shipping and Storage: Steel, 16 and 30 Gallon Capacity
PPP-D-729	- Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)
PPP-P-40	- Preservation and Packing of Hand Tools; Tools and Tool Accessories For Power Driven, Metal and Woodworking Machinery
PPP-T-60	- Tape: Packaging, Waterproof

Federal Standards

FED-STD-101	- Federal Test Method Standard, Test Procedures for Packaging Materials
FED-STD-H28	- Screw-Thread Standard for Federal Services

(Activities outside the Federal Government may obtain copies of federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other federal specifications, standards, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New

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York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

Military Specifications.

MIL-C-104	- Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	- Preservation, Methods of
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-T-152	- Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-H-775	- Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing) And Fittings, Nozzles, and Strainers Packaging Of
MIL-C-3774	- Crates, Wood; Open 12,000- and 16,000- Pound Capacity
MIL-C-5501	- Caps and Plugs, Protective, Dust and Moisture Seal
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, For Preservation and Operation
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of
MIL-I-24092	- Insulating Varnish, Electrical Impregnating, Solvent Containing
MIL-B-26195	- Boxes, Wood-Cleated, Skidded, Load-Bearing Base
MIL-C-52950	- Crates, Wood, Open and Covered

Military Standards.

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-889	- Dissimilar Metals
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing, With Appropriate Test Methods
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packing and Marking

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulations

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Occupation Health and Safety Administration (OSHA) 29 CFR 1910.211-222 Machinery and Machine Guarding

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the federal agency responsible for issuance thereof.)

(Copies of OSHA standards and the standards which are incorporated by reference may be examined at the National Office of the Occupational Safety and Health Administration, U.S. Department of Labor, Washington, DC 20210.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Gear Manufacturers Association (AGMA)

AGMA 360.02	- Manual for Machine Tool Gearing
AGMA 390.03	- Gear Handbook, Volume 1, Gear Classification, Materials, and Measuring Methods for Unassembled Gears

(Application for copies should be addressed to American Gear Manufacturer's Association, One Thomas Circle, Washington, DC 20005.)

American Society for Testing and Materials (ASTM)

ASTM E 380	- Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

International Organization for Standardization (ISO)

ISO - 54	- Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches
ISO - 1122	- Glossary of Gears - Part 1: Geometrical Definitions

(Application for copies should be addressed to the American National Standard Institute, Inc., 1430 Broadway, New York, NY 10018.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG-1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street NW, Washington, D.C. 20037.)

National Fire Protection Association (NFPA), Standard

NFPA-70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

Underwriters Laboratories, Inc. (UL)

UL 987 - Stationary and Fixed Electric Tools

(Application for copies should be addressed to the UL Publications Stock, 333 Pfingsten Road, Northbrook, IL 60662.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one or more complete threading machine(s) for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The threading machine shall be new and of the current design meeting the requirements specified herein and conform to the data shown in Table I. The machine shall be electric motor driven, with the necessary controls. Parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

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TABLE I. Pipe and Bolt Threading Machines

Type	Class	Style	Size	Pipe size (minimum range) Inches	Bolt size (minimum range) Inches	Threading Performance		Motor (minimum) Horsepower
						Pipe size Inches	Time (maximum) Inches	
I	1	A	2	1/8 to 2	1/4 to 2	2	30	1/2
I	1	B	2	1/8 to 2	1/4 to 2	2	30	1/2
I	2	A	2	1/8 to 2	1/4 to 2	2	25	1/2
II	1	A	4	1/2 to 4	5/8 to 3	4	55	3
II	1	B	2	1/4 to 2	1/4 to 2	2	20	3
II	1	B	6	1 to 6	1 to 4	6	160	3
II	2	A	2	1/4 to 2	1/4 to 2	2	20	3
II	2	A	4	1/2 to 4	3/8 to 3	4	35	3
II	2	A	6	1 to 6	1 to 4	6	80	5
II	2	A	8	2 1/2 to 8	2 to 6	8	105	7 1/2

3.2.1 Maintainability. The machine, accessories and adjustable features shall be configured in design to minimize maintenance requirements. Instructions shall provide guidance on normal adjustment, maintenance, repair, and replacement of components.

3.2.2 Descriptions. The narrative descriptions herein are for the convenience of identification and are not intended to preclude the purchase of machines which are otherwise in accordance with the requirements of this specification.

3.2.3 Types. Type I machines are bench mounted and when specified (see 6.2), they shall be furnished with a portable stand. Type II machines are floor mounted.

3.2.4 Classes. Class 1 machines are for light, intermittent duty, and class 2 for heavy, production duty.

3.2.5 Styles. Styles A and B machines shall be capable of machining right-hand threads on standard pipes and bolts within the range specified. When specified (see 6.2), styles A and B machines shall be capable of machining right-hand and left-hand threads on standard pipes and bolts within the range specified. The style A machines shall include cutoff and reaming attachments for pipe within limits of the machine.

3.2.6 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible. Lubrication reservoirs, if applicable, shall have means for checking levels.

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3.2.7 Safety devices. Safety devices shall be provided for parts that present safety hazards to personnel. Safety devices shall not interfere with the operation of the machine or with the operator's performance. The electrical control shall be recessed or otherwise guarded to prevent accidental engagement. The carriage shall be attached to the ways in such a manner as to prevent accidental engagement. The carriage shall be attached to the ways in such a manner as to preclude any possibility of being twisted off the base by torque due to the loosened bolts, way-rails or broken hold-downs. Stops shall prevent the die-head from contacting the chuck (see 6.2).

3.2.7.1 Occupational safety standard. The product shall comply with OSHA standards set forth in 29 CFR 1910.211-222 with respect to machinery protection.

3.2.8 Interchangeability. All replaceable parts shall be manufactured to permit replacement without the need for modification.

3.2.9 Measurement system. The U.S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the threading machines. In this specification, all measurements, dimensions, sizes, and capacities are given in the U.S. system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380.

3.3 Construction. All parts used in fabrication shall be new. The threading machines shall be complete so that when connected to the specified power source, it can be used for any operation for which it is designed. There shall be no process used for reclaiming any part.

3.3.1 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the threading machine industry. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.2 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.3.3 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching.

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3.3.4 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations, however, shall not be employed as repair measures for defective parts.

3.3.5 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.6 Painting. Unless otherwise specified (see 6.2), painting and finishing shall be in accordance with standard commercial practice provided the following minimum criteria is met or exceeded: all surfaces to be painted shall be clean and be free of all foreign matter detrimental to painting and at least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent; (i.e., epoxy enamel), is applied.

3.3.7 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the connections shall be mechanically secured before soldering. Only rosin base fluxes shall be used in soldering operations. Connections to screw type terminals shall be mechanically secured with a means to prevent loss of tightness.

3.3.8 Threads. Threads shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.9 Gears. All gears and pinions used in this machine or its components shall be fabricated to either English (USA) standard inch diametral pitch per American gear Manufacturers Association (AGMA) Standard 360.02 and 390.03 or the Metric per the International System of Units (SI) module ratio per the International Organization for Standardization (ISO) Standards ISO-54, ISO-1122, and ASTM E380.

3.4 Style A, stationary die-head machines. Style A, stationary die-head machines shall hold the work in a revolving spindle and gripping chuck. The machines shall perform machining operations such as threading, cutoff, and reaming. Primarily, the light duty, class 1 machines shall be designed for on-the-job thread cutting, and the heavy duty, class 2 for production work.

3.4.1 Component units. The component units common to stationary die-head machines covered by this specification shall consist of base, headstock and spindle, carriage, stationary die-head, transmission unit, chuck, dies or chasers, cutoff tool, reaming tool, and coolant system.

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3.4.1.1 Base or bed. The base or bed shall be of a suitable metal, either cast or fabricated. The ways shall be of the renewable type securely mounted to the base, or be an integral part of the base. If the ways are not an integral part of the base, they must be restrained as specified in 3.2.7.

3.4.1.2 Carriage. The carriage shall be of a suitable metal, either cast or fabricated, that shall be rigid in construction. It shall be mounted on the ways of the base and be free to move along the ways in either direction from the headstock by manual operation. When specified (see 6.2), a power operated carriage shall be provided. The design shall be such that it supports the die-head and facilitates changing of die-heads.

3.4.1.3 Headstock, spindle, and chuck. The headstock shall be of a suitable metal, either cast or fabricated, constructed to rigidly support and assure alignment of the spindle to the carriage. The bearing(s) supporting the spindle shall be of a size to adequately take the radial and thrust loads at maximum threading and cutting operations. On the forward end of the spindle, a self-centering gripping chuck shall securely grip the pipe or bolt for the machining operations. Unless otherwise specified (see 6.2), a self-centering device shall be provided to support long pipes or bolts. The spindle shall be driven by an electric motor through a transmission at speeds applicable to the size of machine. When specified, a short and close nipple and stud chuck shall be provided. The nipple and stud chuck body shall be provided with drive-chuck grooves or flat surfaces to match the jaw inserts. Wrenches shall be provided as required. The power drive chuck shall incorporate hammer spin-torque tightening. When specified, the universal jaw-wrench-chuck shall be provided, the jaw-wrench shall be self-ejected for safety.

3.4.1.4 Drive and speed. The transmission unit shall drive the spindle at predetermined speeds required for machining operations applicable to the size of the machine specified. The transmission unit may consist of gear or chain drive, or belt, or any combination thereof. Where gear or chain drive is used, an adequate lubrication system shall be furnished. The transmission shall provide for speeds or speed changes and/or the motor shall provide for variable speeds when specified in the contract (see 6.2).

3.4.1.5 Coolant system. The coolant system shall supply coolant to the die-head, cutoff, and reaming tools when in operation. The pump shall be mounted to the machine and may be driven by an individual motor or by the transmission unit. The control valve(s) shall be located on the machine convenient to the operator. The coolant spray shall be directed on the opposite side of the die-head than that of the open end of the pipe being threaded. This is to prevent the oil spraying inside the pipe. The coolant spray shall be controllable toward both the chuck and die-head. A pressure relief valve shall be incorporated within the system. A filter shall be

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provided between the pickup screen and the pump and shall filter out all metal cuttings and foreign particles that may damage the pump and dies. The filter shall be cleanable.

3.4.1.6 Die-head. The die-head shall be the quick-opening type with opening and closing movements controlled manually. When specified (see 6.2), automatic operation shall be provided. The die-head shall accommodate the range of die chaser sizes specified for the machine and shall have adequate adjustment for oversize and undersize threads. The quick, and quick-self opening die-head shall employ operator handle tightener for die adjustments unless other types are specified in the contract. The index marks that indicate die adjustment shall be marked on the outer periphery of the die-head and adjuster ring. The swinging end of the die-head frame shall rest in a slot or against a backup block to prevent lateral movement by thrust pressure when starting threads.

3.4.1.6.1 Dies or chasers. Dies or chasers shall be either high carbon steel, or high speed steel and hardened to Rockwell C57 to 68. When specified (see 6.2), metric dies with appropriate die-heads shall be provided for cutting metric (SI) threads on pipe and bolts. Each machine shall be furnished with complete sets of right-hand dies or chasers in the Unified Coarse Thread UNC Series and one complete set of right-hand dies or chasers in the American National Pipe Thread NPT Series, both to cut threads in accordance with FED-STD-H28 and within the range of the machine as specified in Table I. Dies or chasers other than those indicated above shall be as specified (see 6.2).

3.4.1.7 Cutoff tool. A cutoff tool shall be furnished and shall be capable of cutting off pipe within the size limits of the machine specified.

3.4.1.8 Reamer. A reamer or cutter shall be furnished on all stationary die-head machines for the purpose of reaming pipe within the size limits of the machine specified. The free swinging end of the reamer frame shall fit in a slot or against a backup block to prevent lateral movement by thrust pressure during reaming operations.

3.5 Style B, revolving die-head machines. Style B revolving die-head machines shall hold the work in a stationary vise. It shall be suitable for operations such as threading bolts, studs, and pipe nipples, and pipe or bolt threading after bending.

3.5.1 Component units. The component units common to revolving die-head machines shall consist essentially of base, headstock and spindle, carriage, revolving die-head, transmission unit, vise or chuck, dies or chasers, and coolant system.

3.5.1.1 Base or bed. The base or bed shall meet the requirements specified for style A stationary die-head machines (see 3.4.1.1).

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3.5.1.2 Carriage. The carriage shall be of a suitable metal, either cast or fabricated, that shall be rigid in construction. It shall be mounted on the ways of the base and be free to move along ways in either direction from the headstock by manual operation. When specified (see 6.2), a power operated carriage shall be provided. The design shall be such that it shall support a vise or chuck for holding work in position for the threading operation.

3.5.1.3 Vise or chuck. A vise or chuck, as specified (see 6.2), shall be furnished for holding the pipe or bolt to be treated. The vise or chuck shall be manually or power operated, as specified (see 6.2). It shall be securely mounted on the carriage with provision for alignment with the revolving die-head.

3.5.1.4 Headstock or spindle. The headstock shall be of a suitable metal, either cast or fabricated, constructed to support the spindle and assure alignment of the spindle to the carriage. The bearing supporting the spindle shall be of a size to adequately take the radial and thrust loads at maximum threading and cutting operations. The revolving die-head shall be securely mounted on the forward end of the spindle. The spindle shall be driven by an electric motor through a transmission unit at speeds applicable to the size of the machine.

3.5.1.5 Transmission unit. The transmission unit shall meet the requirements specified for style A (see 3.4.1.4).

3.5.1.6 Die-head. The die-head shall meet the requirements specified for style A (see 3.4.1.6), and shall be manual or automatic operation, as specified (see 6.2).

3.5.1.6.1 Dies or chasers. Dies or chasers shall be either high carbon steel, or high speed steel and hardened to Rockwell C57 to 68. Each machine shall be furnished with complete sets of right-hand dies or chasers in the UNC Series and one complete set of right-hand dies or chasers in the NPT Series, both to cut threads in accordance with FED-STD-H28 and within the range of the machine as specified in Table I. Dies or chasers other than those indicated above, including metric sizes, shall be as specified (see 6.2).

3.5.1.7 Coolant system. The coolant system shall meet the requirements specified for style A (see 3.4.1.5).

3.6 Electrical equipment.

3.6.1 Motor(s). Unless otherwise specified (see 6.2), motor(s) shall be universal in accordance with NEMA MG-1.

3.6.2 Voltage and frequency. The operating voltage and frequency characteristics shall be as specified by the procuring agency (see 6.2).

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3.6.3 Power supply cord. Unless otherwise specified (see 6.2), each type I threading machine shall be furnished with 8 to 10 feet of three-conductor flexible power cord, type SO, STO, or SJO hard service in accordance with NFPA-70. The cord shall be provided with a three-prong polarized grounding connector conforming to W-C-596.

3.7 Fungus control. When specified (see 6.2), electrical components and circuit elements shall be treated for fungus resistance in accordance with type II requirements of MIL-T-152 except that the statement of treatment is not required. Windings of the electric motors shall be given two impregnating coats of varnish conforming to the requirements of MIL-I-24092, type M, class 130 through 220 (manufacturer's option).

3.8 Identification marking. Unless otherwise specified (see 6.2), each threading machine shall be provided with a suitable nameplate which shall be permanently marked and securely fastened to an integral part of the frame or the housing in a conspicuous location. The nameplate shall be in accordance with the manufacturers standard practice and shall enumerate as a minimum that information required by UL 987.

3.9 Technical data. Data shall be furnished as specified (see 6.2.1).

3.10 Workmanship. Workmanship of the threading machine and accessories shall be of the best quality prevailing among manufacturers of this type of equipment. The items shall be free from all imperfections which will adversely affect the general appearance, function, and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1. Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products submitted to the government for acceptance comply

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with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item noted in 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests under 4.7 through 4.9. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production units being offered for acceptance under the contract. The inspection shall consist of (a) through (e) as follows and failure of any unit to pass an inspection or test shall be cause for rejection of the unit.

(a) Product examination (see 4.6).
(b) Threading performance test (see 4.7).
(c) Hardness test (see 4.8)
(d) Packaging validation test (see 4.9).
(e) Packaging inspection (see 4.11).

4.5 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or Test	Lot or Batch Size	Samples Size
Product examination (see 4.6)	1-90	100%
Threading performance test (see 4.7)	1-90	13
Hardness test (see 4.8)	1-90	13
Packaging validation test (see 4.9).	1/	-
Packaging inspection (see 4.11).	1-90	3

The lots shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size, 100% inspection is required. Each lot shall be accepted with no defects and rejected if one or more defects are found.

1/ The packaging validation tests shall be conducted on one of the packaged items selected from the first lot submitted for quality conformance inspection and tests.

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4.6 Product examination. Visually, dimensionally and manually examine each threading machine to determine conformance with the requirements of 3.2 through 3.10. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examination includes measuring dimensions as specified. Manual examinations shall include the operation of moveable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the threading machine and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.2.8), shall be cause for rejection.

4.7 Threading performance test. The machine shall be tested by machining standard bolt and pipe threads in the size range specified in Table I. One sample of each thread in size increments of 1/4 inch, for bolts and in the applicable increments of pipe threads shall be produced. The resulting thread shall be checked for conformance to FED-STD-H28.

4.8 Hardness test. A certificate of conformance (COC) shall be required verify that the following parts meet the hardness required below. Paragraph number are referenced from Product Engineering Requirements.

PART DESCRIPTION	HARDNESS
Dies or chasers (see 3.4.1.6.1)	C-57 to C-68
Dies or chasers (see 3.4.1.6.1)	C-57 to C-68

4.9 Packaging validation tests. After the Quality Assurance inspection and tests have been successfully completed, one threading machine, packaged in accordance with the requirements of Section 5 Packaging, shall be selected at random and subjected to the following rough handling tests specified in FED-STD-101. If a First Article Test (FAT) is required by the contract, the packaging validation test shall be conducted on the threading machines used for the FAT. If the FAT is not required by the contract, then a threading machine selected from the production lot used for the test in paragraph 4.7 shall be used for the packaging validation test. The apparatus used shall be as described in FED-STD-101. A dummy load shall not be used. All test are to be conducted under ambient conditions. Environmental temperature/humidity tests are not required. After the test has been completed, the container shall be inspected. Defects in accordance with those described below shall be cause for rejection. The unit shall be unpacked and again subjected to the Quality Assurance inspections and test of paragraph 4.7 to reaffirm that the

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threading machine meets the requirements specified in Section 3. After all requirements have been met, the unit shall be inspected and repackaged (see 4.11) before submission for government acceptance. After testing, any damage to the packaging shall be replaced prior to acceptance of the lot.

4.9.1 Mechanical handling test. Test shall be conducted using Method 5011.1 Paragraph 6.2 (Lifting and transportation by forklift truck) of FED-STD-101. Criteria for passing/failing of the test stated in Paragraph 7.1.5 of Method 5011.1.

4.10 Evidence of compliance with OSHA standards. Evidence of compliance with applicable safety and health requirements shall be reviewed (see 3.2.7.1 and 6.3).

4.11 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 First article. When specified (see 6.2), the supplier shall furnish one complete pack for first article inspection. Whenever first article inspection is specified or performed, unit pack examination and tests for packaging may be performed also (see paragraphs 4.2 and 5.10). The first article may be either a preproduction or an initial production pack which conforms to the requirements of this specification. In either case, the approved first article and the production packs shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment and packaging in accordance with the requirements of this specification

5.2 Disassembly. When practical, for the protection of components, attachments and accessories against damage, pilferage, or to reduce cubage, items may be disassembled only to a necessary basic degree. Disassembled parts shall be preserved, anchored, braced, blocked, and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special skills or tools.

5.3 Disassembled parts marking. Disassembled parts shall be marked, when necessary with instructions, to facilitate reassembly. Removed parts and mating parts on the item shall be marked by stenciling, on the part, or by use of tags. Tags shall be waterproof and ink used (on parts or tags) shall be waterproof. Tags shall be firmly secured to each individual part.

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5.4 Materials. Materials used in preservation/packaging and packing of machinery shall be as specified herein. Materials shall be new and free from all defects and imperfections which may affect the serviceability of the completed package or pack.

5.5 Processing conditions. Preservation/packaging shall be accomplished within buildings which have facilities to limit the accumulation of dust and moisture on the equipment being processed. Machinery brought into processing areas from outside shall be allowed sufficient time to reach temperature before processing. Cleaning, drying, and application of preservatives shall progress in an interrupted series of operations. When periods of interruption are necessary, temporary protection shall be provided to partially processed items, as required, to insure against contamination.

5.6 General requirements. Components parts of machinery, accessories, and maintenance tools shall be cleaned, dried, preserved, packaged, and packed as specified herein. Parts and assemblies which are not covered by this specification shall be preserved and packaged utilizing the materials and methods specified for similar parts and assemblies described herein. Cleaning and drying procedures and preservatives and their application, specified herein, shall conform to the criteria and guidelines of MIL-P-116.

5.7 Preservation/packaging. Preservation shall be Level A or C, as specified in the contract (see 6.2.1).

5.7.1 Level A.

5.7.1.1 Preservation/packaging. unless otherwise specified (see 5.7.1.28), the basic machine shall be preserved and packaged in accordance with method I criterion of MIL-P-116. Electronic and electrical equipment, hand tools and accessories shall be preserved and packaged as specified herein. When it has been determined that a preservative is required to protect a specific surface or item, but has not been specified herein, the preservative utilized shall be selected from MIL-P-116. All P type preservatives specified herein shall be IAW the requirement specified in MIL-P-116.

5.7.1.2 Cleaning and drying. Thorough cleaning and drying of all bare ferrous metal surfaces is essential before the application of a preservative or other protective finish such as paint, plating, etc. Care must be taken to limit the handling of cleaned parts prior to the assembly of new machinery. Where practicable, processors and assemblers shall wear rubber or plastic gloves while handling cleaned parts.

5.7.1.3 Internal surfaces. All ferrous metal surfaces of parts which will be utilized internally in the assembly of new machinery shall be cleaned in accordance with process C-5 and dried in accordance with the applicable procedure of MIL-P-116. In the interim, to preclude contamination from

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forming on cleaned surfaces prior to the assembly operation, parts shall be coated with a light weight preservative oil conforming to P-9. To prevent dust from accumulating on cleaned parts, cover with a sheet of greaseproof material, such as MIL-B-121. In the event the supplier deems it necessary to clean a particular mechanism after draining the operational lubricant, flush with a solution of one part lubricating oil P-10, type I, grade 30 and nine parts cleaning solvent conforming to P-D-680. Otherwise, the mechanism need not be cleaned. All ferrous metal internal surfaces shall be coated with the lubricant or a preservative. Internal mechanisms shall be cycled to assure all areas are protected with a coating of the lubricant before either sealing or draining. Any unprotected areas will have a preservative coating applied.

5.7.1.4 External surfaces. All accessories and attachments such as chucks, faceplates, steady rests, toolholders, and abrasive wheels shall be removed from the assembly prior to the external cleaning operation. All external critical ferrous metal surfaces (fit or function) of the machine shall be cleaned in accordance with process D-5 and dried in accordance with the applicable procedures of MIL-P-116. Noncritical ferrous metal surfaces shall be cleaned in accordance with process C-3 and dried as specified above. Nonmetallic composites and painted metal surfaces shall be cleaned by wiping with a clean, dry cloth. (Care must be taken to prevent cleaning solvents from coming in contact with nonmetallic composites.) When cleaned surfaces of the machine show signs of rusting before application of the final preservative, the cleaning process shall be confined to those specific small areas of the assembly. Alternatively, the solvent used for the final wiping operation shall contain a small amount (approximately 5 percent by volume) preservative oil conforming to P-10, type 1, grade 30. (It is extremely important that elapsed time between operations be held to an absolute minimum.)

5.7.1.5 Disassembled components. Parts detached from the basic machine shall be cleaned and dried as specified in 5.7.1.2, 5.7.1.3, and 5.7.1.4.

5.7.1.6 Steam and air lines. Drain all water accumulation and dry by injecting moisture-free air into the system.

5.7.1.7 Maintenance or machine tools and tool accessories. All machine, maintenance, metal or woodworking, hand tools and any accessories thereof shall be preserved and packaged in accordance with Level A requirement of PPP-P-40.

5.7.1.8 Electrical equipment. Electrical equipment, whether attached or detached, shall be cleaned, dried and preserved in accordance with Level A requirements of MIL-E-16298 or MIL-E-17555, whichever applies to the individual electrical item.

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5.7.1.9 Electrical motors. Electric motors, whether attached or detached, shall be preserved and packaged in accordance with Level A requirements of MIL-E-16298, and applicable subparagraph for assembled machines.

5.7.1.10 Electrical control devices. Control units, whether attached or detached, shall be preserved and packaged in accordance with Level A requirements of MIL-E-17555.

5.7.1.11 Electrical wiring system. Exposed ends of wires, plugs openings, sockets, connector plugs, terminals and openings into switches and junction boxes shall be sealed with tape or cap-plugs conforming to PPP-T-60, type IV or MIL-C-5501, respectively. Electric cable shall be coiled to a minimum safe diameter and secure with cotton twine or tape to prevent uncoiling.

5.7.1.12 Hoses and fittings. Hoses and fittings which are not attached to the basic machine shall be cleaned, dried and preserved in accordance with Level A requirements of MIL-H-775.

5.7.1.13 Lubricating systems. The machine, with all lubricants, greases, and other lubricating compounds either installed or included (see 5.8.1.9) shall have all vent openings in the lubrication systems temporarily capped, plugged or taped closed to prevent loss of lubricants during shipment and storage. Tags, warning that "ALL TEMPORARY CAPS, PLUGS, OR TAPE MUST BE REMOVED BEFORE START-UP OR USE OF THE ITEM", shall be attached to all temporary caps, plugs, or tape intended to be vented to the atmosphere during operation. In the event a lubrication reservoir cannot be covered and sealed, it shall appropriately be preserved and the lubricant shall be provided in a sealed container and shipped with the machine (see 5.8.1.9). A warning tag shall be placed in a conspicuous place near the filling area, warning the user to "INSTALL THE FURNISHED LUBRICANT BEFORE OPERATION OF THE EQUIPMENT".

5.7.1.14 Coolant systems. Coolant systems shall be drained and cleaned of all coolant residue and shall be preserved by flushing with P-3 preservative compound leaving a residual film of the preservative in the coolant lines and tank. Drain off the preservative and close the systems. Interior surfaces of the tank/reservoir which have not been covered or flooded with preservative while the machine is in operation shall be manually coated by brushing or spraying with similar type preservation.

5.7.1.15 High-speed spindle bearings. Reservoir shall be filled to normal operating level with manufacturer's lubricant and sealed closed for shipment and storage. If the system is of such design that the lubricant will leak out during shipment and storage, it shall be drained. The system shall then be preserved by filling with preservative oil P-10, type I, grade 30, drained, sealed, and covered.

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5.7.1.16 Hydraulic systems. Both open- and closed-type systems shall be filled to normal operating level with either the manufacturer's operative hydraulic oil or hydraulic oil conforming to MIL-H-6083. After filling, the machine shall be cycled to circulate the oil through the entire system. Prior to disconnecting the machine from the power source, place the piston(s), except opposing piston, in their innermost position. The oil shall remain in the reservoir, providing the reservoir can be sealed to prevent any leakage during shipment and storage. If it cannot be sealed to prevent leaking, it shall be drained and preserved and the hydraulic oil shall be provided in a sealed container and shipped with each machine.

5.7.1.16.1 Hydraulic systems for tracer mechanism. These systems shall be filled to an operating level with manufacturer's hydraulic oil or MIL-H-6083 and sealed or preserved as stated above in Hydraulic systems paragraph.

5.7.1.17 Grease fittings. Charge all grease fittings with manufacturer's compound or with preservative compound conforming to P-11.

5.7.1.18 Nonlubricated interior surfaces. Surfaces of internal gears which do not run in lubricant shall be thoroughly coated with P-7 preservative.

5.7.1.19 Air motors. Interior surfaces of air motors and air lines shall be coated with a preservative oil conforming to P-10, type I, grade 30 by injecting oil into the air stream while operating the machine until the oil appears at the exhaust ports. Air inlets and outlets shall be sealed with plastics caps, plugs, or tape conforming to MIL-C-5501 or PPP-T-60, type IV, respectively. Air cylinders shall not be disassembled from the machine unless parts are vulnerable to damage during shipment and storage.

5.7.1.20 Gear cases. Fill the gear cases with normal lubricating oil. Shift gears into all possible positions while operating the machine. Make sure all surfaces are coated with lubricating oil and close and seal all openings to prevent any leakage during shipment and storage.

5.7.1.21 Sliding or friction surfaces. Surfaces of ways, gibs, driving gears, traversing, and adjusting screws, and other critical, exterior operating machined surfaces shall be coated with P-2 preservative compound. Parts of the machine which are adjustable shall be locked into place by means provided and blocked to assure movement shall be kept to a minimum.

5.7.1.22 Non-critical surfaces. Exterior, non-critical, ferrous metal surfaces of the basic machine which have not been painted or plated, shall be coated with type P-2 preservative and covered. Such places as legs, bolts, etc., from which the preservative need not be removed may be preserved with P-1 or P-19.

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5.7.1.23 Bearings and journals. Grease lubricated bearings shall be preserved with type P-11 preservative. Oil lubricated journals shall be filled with normal oil lubricant or with lubricating oil P-10, type I, grade optional.

5.7.1.24 Drive belts and pulleys. Drive belts shall either be removed from the machine or released from tension. Belts removed from the machine shall be packaged in accordance with method IC-I criterion of MIL-P-116. Ferrous metal pulleys shall be coated with a rust-inhibiting primer conforming to TT-P-664.

5.7.1.25 Closed dial indicators. Closed dial indicators shall be covered with a minimum of 1 inch thick uncompressed cushioning materials conforming to PPP-C-843, type II, class B. Secure cushioning material in place with tape conforming to PPP-T-60, type IV. Other type cushioning materials and tapes may be used provided the materials are water-resistant, non-corrosive, and will provide equivalent or better protection than specified above.

5.7.1.26 Drawers and door assemblies. Sliding surfaces of drawer guides shall be coated with P-11 preservative grease. Interior surfaces of door hinges shall be lubricated by the application of P-10, grade 30 preservative oil. In addition to mechanical locks or catches incorporated in the machine, secure doors and drawers with metal strapping or other devices which are suitable for the purpose intended. Fiberboard pads shall be placed under metal strapping, to prevent marring the painted surfaces.

5.7.1.27 Closure of openings. Small openings (except vents and louvers installed for ventilation purposes) which will admit dust or water shall be sealed with tape conforming to PPP-T-60, type IV. Large openings shall be covered with waterproof barrier material conforming to class E-1 or L-4 of PPP-B-1055, or as a minimum, plastic material conforming to L-P-378, type and class optional. Secure materials in place with tape specified above. When very large openings are covered, or when location of openings render covering vulnerable to puncture, coverings shall be protected by applying wood, plywood or sheet metal over the materials. Open ends of all piping and fittings shall be plugged to prevent the entrance of foreign materials, using pipe fittings, plastic cap-plugs conforming to MIL-C-5501 or tape conforming to above.

5.7.1.28 Technical data. Technical data furnished with the machine shall be packaged in accordance with method IC-I criterion of MIL-P-116. The package shall be marked "TECHNICAL DATA" in bold black letters. The package shall be protected against puncture and abrasion. The data shall be either attached to the machine with tape conforming to PPP-T-60, type IV, placed in a storage compartment of the machine, or positioned in a consolidated box containing accessories or disassembled components. If placed inside an area, that area shall be tagged or marked "TECHNICAL DATA ENCLOSED" so as to be readily available for use.

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5.7.1.29 Method II packaging. When specified (see 6.2.1), complex items which cannot be treated entirely externally with a contact preservative or where removal of the preservative would prevent timely use of the machine shall be packaged in accordance with Method II criterion of MIL-P-116. Accessories, maintenance tools, control units, and all internal surfaces of the machine shall be preserved and packaged as specified herein. Method II packs will bear precautionary labels as stated in MIL-STD-129. Upon receipt of a Method II package, it shall be inspected to assure the vacuum seal is intact before being placed in storage or shipped elsewhere.

5.7.1.30 Protection of preservatives. All exposed preserved surfaces of machines which are subject to being disturbed during the packing operation shall be loosely covered with barrier material conforming to MIL-B-121, type I, grade A. Not less than two layers of similar-type barrier material shall be used to separate wood or metal blocking and finished metal surfaces of the machine. When considered necessary, secure barrier in place with tape conforming to PPP-T-60, type IV.

5.7.1.31 Consolidated packaging. When size and weight permit, packaged accessories, maintenance tools, and small disassembled components shall be placed in drawers and stowage compartments of the machine. When it is not feasible to stow parts as specified above, place parts in a box(s) conforming to PPP-B-636, class WR or WWVR. Large bulky, disassembled components, depending on size and weight, may have to be packed in box(s) conforming to PPP-B-601, domestic type or PPP-B-621, class 1 rather than PPP-B-636.

5.7.1.32 Cushioning. Cushioning materials and methods of application shall conform to the criteria and guidelines of MIL-P-116 and MIL-STD-1186. In the event there are conflicting requirements, MIL-STD-1186 will have precedence.

5.7.1.33 Movable parts. Operating heads, sliding tables, rams, counterbalances, and other movable parts shall be secured by mechanical devices incorporated in the machine. The devices alone do not always provide adequate support to prevent movement in the handling and transporting of the item and therefore must be augmented with appropriate wood blocking, metal tie rods used in conjunction with wood blocking, metal strapping, or other devices which will immobilize the part. Handwheels and levers, and similar controls shall be secured in position with common annealed, wire or steel strapping. Fiberbeard pads shall be used under strapping, ets, to prevent marring.

5.7.1.34 Positioning assemblies. Assemblies having vertical movement shall be blocked and braced in a lowered position; those capable of moving traversely shall be centered for blocking and bracing.

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5.7.1.34.1 Tables. Tables and other components moving on ball bearings or other types of high efficiency, low-friction ball or roller bearing assemblies shall be removed, or blocked, and all components treated in such a manner that neither the way(s), surface(s), nor the anti-friction devices will be subject to brinelling or other damage.

5.7.1.34.2 Counterweights. Counterweights shall be locked and blocked in a manner which will release all tension from supporting cables, chains, and springs. Concealed counterweights suspended in closed columns shall be removed or blocked to prevent vertical or horizontal movement.

5.7.2 Level C preservation. Preservation of metal and woodworking machinery shall conform to the requirements of MIL-STD-1190.

5.8 Packing. Packing shall be Level A, B, or C, as specified in the contract.

5.8.1 Level A.

5.8.1.1 Machinery weighing less than 2,500 pounds. Unless otherwise specified (see 6.2), machinery including accessories, weighing less than 2,500 pounds, shall be packed in cleated plywood boxes conforming to PPP-B-601, overseas type; nailed wood boxes conforming to PPP-B-621, class 2, style 2, 3, 4, or 7; skidded boxes conforming to MIL-B-26195, type II, style A, class optional, with plywood superstructure. Selection of containers shall be governed by the size and weight limitations of the applicable container specification. Machines designed with mounting holes in areas that can withstand rough handling without breakage shall be bolted to an auxiliary base before being packed in a box conforming to PPP-B-601 or PPP-B621. Contents of boxes shall be arranged, anchored, blocked, braced, and cushioned in accordance with MIL-P-116 and MIL-STD-1186. In the event there are conflicting requirements, MIL-STD-1186 will have precedence. Top-heavy machines shall be packed as specified in 5.8.1.5. Box closure and strapping requirements shall be in accordance with the appendix to the applicable box specification.

5.8.1.2 Machinery weighing 2,500 to 30,000 pounds. Unless otherwise specified (see 6.2.1), machines weighing 2,500 to 30,000 pounds shall be packed individually in crates conforming to MIL-C-104, type I, class 2, style A. To conserve cubage, safeguard components of the machine, and to comply with weight and dimensional limitations of the crate specification, subassemblies removed through disassembly and control units may be packed in separate shipping containers. When packed separately, items weighing less than 2,500 pounds shall be packed as specified in 5.8.1.1. Arrangement of contents, anchoring, blocking, bracing, and cushioning shall be in accordance with the appendix to the crate specification, MIL-STD-1186, and as specified

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herein. MIL-STD-1186 will have precedence when there are conflicting requirements. When practical, to facilitate loading into common carriers, the height of the crated machine shall be limited to 9 feet 8 inches.

5.8.1.3 Buttress blocking. Hold down bolts on machines weighting over 8,000 pounds shall be reinforced with backup buttress extending from the machine base to the skid header; blocking shall be parallel to skid runners and in line with bolt holes on the machine. The blocks shall be similar to that used for the skid header. Carriage or step bolts shall be utilized to secure the block to the skid base.

5.8.1.4 Unsupported columns. Unsupported columns of the machine shall be reinforced by placing snug-fitting blocks between the columns and the skid platform or machine base, as applicable. Blocks extending from the column to the skid base shall be fitted so that tightening of the hold down bolts will not place undue stress on the column(s).

5.8.1.5 Top-heavy machinery. To safeguard equipment and personnel, top-heavy machine tools may be packed in a horizontal position, design permitting. Blocking and bracing shall be located at points of the machine with sufficient rigidity to carry the machine weight. Crates for top-heavy items requiring upright skidding due to design, structural weakness, etc., shall be sized to provide stability in handling and transit. The length and width of the skid platform shall extend beyond the machine a distance equivalent to the height of the center of gravity of the machine above the base.

5.8.1.6 Leg-type machines. Leg-type machines with slotting holes for mounting shall be reinforced by placing a nominal 2- by 4-inch wood block against the leg opposite the slot opening and nailing to the skid base.

5.8.1.7 Bolting. Machines shall be secured to the skid base by not less than four bolts. When sufficient bolting points are not available, J-clamps or tie rod assemblies as specified in 5.7.1.33, or other devices, shall be used to provide the minimum of four holding points. Bolts shall pass completely through the skid runner or sleeper, as applicable, and the base flooring.

5.8.1.8 Consolidated containers. Consolidated containers specified in 5.7.1.31 shall be placed inside the container housing the machine. The boxes shall be restrained by the use of metal strapping of sufficient strength or tie rod assemblies.

5.8.1.9 Operational oils. Operational oils furnished with the machine, but not placed in the machine stowage receptacles, shall be stowed in metal drum(s) conforming to PPP-D-705, type I or II, or PPP-D-729, type I or II. Drums shall be positioned upright inside the shipping container housing the machine. Sufficient blocking shall be applied to immobilize the drum(s)

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inside the shipping container. Alternatively, when the drum(s) tend to increase the cubage of the shipping container, the drum(s) may be shipped as individual containers.

5.8.1.10 Machines exceeding crate limitations. Unless otherwise specified (see 6.2.1), machines exceeding the weight and size limitations of MIL-C-104 shall be packed in a similarly designed crate. The crate's structural members and anchoring devices shall be increased in number and size as needed to accommodate the load to be packed therein.

5.8.2 Level B. Machines weighing 16,000 pounds or less shall be packed for shipment utilizing boxes conforming to PPP-B-601, domestic type, style A, B, I, or J; PPP-B-621, class 1, style 2, 3, 4, or 7; MIL-B-26195, type I, style A; or crates conforming to MIL-C-52950 or MIL-C-3774. Selection of the container shall be governed by the size and weight limitations of the applicable container specification. Machines designed with mounting holes in areas that can withstand rough handling without breakage shall be bolted to an auxiliary base before being packed in a box conforming to PPP-B-601 or PPP-B-621. Arrangement of contents, anchoring, cushioning, blocking, bracing, and container closure shall be the same as for level A (see 5.8.2). Items packed in open crates shall be covered with a waterproof shroud conforming to MIL-STD-1186. The shroud shall be secured to the base of the machine. Polyvinyl chloride (PVC) and ethylene vinyl acetate (EVA) shall not be acceptable as an intimate wrap or shroud material due to the possible corrosive effect of PVC or EVA vapors on the shrouded item and its component parts. Machines exceeding 16,000 pounds shall be packed as specified in 5.8.1.2.

5.8.3 Level C packing. Packing of metal and woodworking machinery shall conform to the requirements of MIL-STD-1190. When authorized by the transportation media and when safe handling with mechanical equipment is provided, the machines may be mounted on skids or a skid base in lieu of being boxed or crated. Openings in skids shall be a minimum of 3 inches in height to provide access for forklift tines. Machines not packed in fully sheathed containers shall be completely covered with a waterproof plastic shroud. The shroud shall be secured to the base of the machine. Polyvinyl chloride (PVC) and ethylene vinyl acetate (EVA) shall not be acceptable as an intimate wrap or shroud material due to the possible corrosive effect of PVC or EVA vapors on the shrouded item and its component parts.

5.9 Marking.

5.9.1 Military marking. Military marking shall be in accordance with MIL-STD-129. Bar code markings are required and shall be in accordance with MIL-STD-129. Center of balance markings and arrow markings indicating THIS SIDE UP, etc shall be applied to each shipping container for all levels for the protection of the machinery and personnel.

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5.9.2 Civil agencies. Shipment to civil agencies shall be marked in accordance with FED-STD-123.

5.10 Rough handling. The completed unit packs, shipping container(s) and all contents shall be cabable of withstanding the shipping and handling conditions expected from fork lift handling and truck transport. The container, its contents and the packaging shall not be damaged, functionally or physically, or items displaced by the appropriate rough handling tests specified.

5.11 Workmanship. Workmanship shall be such that when the proper procedure is followed, materials and equipment being processed will be provided the maximum protection against corrosion and deterioration consistent with the shipment hazards known or anticipated, and suitable for the term of storage expected.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended for threading pipe, tubing, nipples, and bolts. Where applicable, attachments for cutoff and reaming are included.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type, class, style, and size, as required (see 1.2.1).
(c) First article, when required (see 3.1).
(d) Portable stand, when required (see 3.2.3).
(e) Right-hand and left-hand capabilities, when required (see 3.2.5).
(f) Special safety devices, when required (see 3.2.7).
(g) Painting, if different (see 3.3.6).
(h) Power operated carriage, when required (see 3.4.1.2).
(i) Self-centering device (style A), if different (see 3.4.1.3).
(j) Speed changes, as required (see 3.4.1.4).
(k) Automatic operation, when required (See 3.4.1.6).
(l) Other dies or chasers (style A), as required (see 3.4.1.6.1).
(m) Power operated carriage, when required (see 3.5.1.2).
(n) Chuck or vise (style B), as required (see 3.5.1.3).
(o) Chuck/vise (style B) manual or power operation as required (see 3.4.1.6.1).
(p) Die-head (style B) manual or automatic operation, as required (See 3.5.1.6).
(q) Other dies or chasers (style B), as required (see 3.5.1.6.1).
(r) Motor(s), if different (see 3.6.1).
(s) Voltage and frequency, as required (see 3.6.2).
(t) Power supply cord and connector, if different (see 3.6.3).

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(u) Fungus control, when required (see 3.7).
(v) Marking, if different (see 3.8).
(w) First article inspection, if different (see 4.3).
(x) Quality conformance inspection, if different (see 4.4).
(y) Level of preservation, packaging, and marking, required (see 5.7 and 5.8).

6.2.1 Contract data requirements. Required technical data, such as operators manuals, parts lists, foundation and anchor bolt plans, wiring diagrams, and other instruction for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in to the contract (see 3.9).

6.3 OSHA standards. Products which meet all requirements of this specification comply with the OSHA standards published prior to the date of this specification and identified in section 3 (see 3.2.7.1).

MILITARY INTERESTS:

Custodians
Army - AL
Navy - YD

Review Activity
DLA - GS

User Activites
Navy - MC, SH

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

Army - AL

DoD Project 3419-0221

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